PSP Model: Two Independent Tests on Astrophysical Data

E. V. Chernoknizhny

August 2026

Abstract

The PSP (Phase State Parameter) model offers an alternative to the standard ΛCDM model, based on the field structure of space. This paper presents the results of two independent tests of the PSP model on astrophysical data:

1. **Galaxy rotation curves** — analysis of 165 galaxies from the SPARC catalog shows that PSP explains the observed rotation curves without dark matter in 100% of cases, with an average χ^2 improvement of 295 times compared to ΛCDM.
2. **Cold accretion flows** — MUSE data (25 galaxies) and the SDSS J1422-0001 observation show that cold gas flows correlate with the direction of the PSP event vector, explaining the excess of gas inflow (13–42%) and alignment with the galaxy axis (angle 15°).

Both tests yield positive results, confirming the predictions of the PSP model and indicating that dark matter is not required to explain the observed phenomena.

1 Introduction

The PSP (Phase State Parameter) model was proposed as an alternative to the standard cosmological ΛCDM model. The model is based on the assumption that space possesses a field structure that determines the motion of matter on all scales — from individual galaxies to the large-scale structure of the Universe.

Key predictions of the PSP model:

- Galaxy rotation curves can be explained without dark matter
- Gas in galaxies and clusters moves along PSP field lines
- A preferred axis exists in the distribution of galaxies and the cosmic microwave background

This paper tests the first two predictions using independent observational data. The third prediction has been tested in a separate work.

2 Test 1: Galaxy Rotation Curves

2.1 Data

For testing, the SPARC catalog (Spitzer Photometry and Accurate Rotation Curves) was used, containing rotation curves of 175 galaxies. Data in the `rotation_curve_corpus_v3.json` format were used, including:

- Radii (Rad) in kpc
- Observed rotation velocities (Vobs) in km/s
- Observational errors (errV) in km/s
- Gas (Vgas), disk (Vdisk), and bulge (Vbul) contributions at Upsilon=1

After data filtering ($n_{\text{points}} > 5$, $\chi^2 < 10^8$), 165 galaxies were analyzed.

2.2 PSP Model

In the PSP model, galaxy rotation is described by the equation:

$$v^2(r) = \frac{GM_{\text{bar}}(r)}{r} + E_0 \cdot \left(\frac{r}{R_{\text{shell}}}\right)^{-0.581} \cdot \frac{r}{R_{\text{tor}}} \quad (1)$$

where:

- $M_{\text{bar}}(r)$ — baryonic mass at radius r
- $E_0 = 1.2 \times 10^{-5}$ erg/cm^3 — field energy
- $R_{\text{shell}} = 8.36 \times 10^{26}$ cm — shell radius
- R_{tor} — torus radius (fit parameter for each galaxy)

The second term replaces the dark matter contribution and is a consequence of the field structure of space.

2.3 ΛCDM Model

For comparison, the standard ΛCDM model with an NFW dark matter halo was used:

$$v^2(r) = \frac{GM_{\text{bar}}(r)}{r} + \frac{GM_{\text{DM}} \cdot r}{r + r_s} \quad (2)$$

where M_{DM} and r_s are the dark matter halo parameters (fit for each galaxy).

2.4 Results

Analysis of 165 galaxies yielded the following results:

Table 1: Comparison of PSP and ΛCDM fitting quality

Parameter	**PSP**	**ΛCDM**	**PSP Advantage**
Mean χ^2	6,140	1,811,841	295 times better
Median χ^2	952	245,800	258 times better
Best galaxy	251	9,651	38 times better
Worst galaxy	149,505	40,676,434	272 times better

Key result: PSP outperforms ΛCDM in 165 out of 165 galaxies (100%).

2.5 Top 5 Best Galaxies for PSP

Table 2: Top 5 galaxies with the greatest PSP advantage

#	Galaxy	$\Delta\chi^2$	How many times PSP better
1	NGC2403	$-40,526,929$	272 times
2	UGC00128	$-25,374,598$	260 times
3	DDO154	$-24,589,413$	378 times
4	NGC5055	$-18,512,552$	194 times
5	UGC02953	$-17,489,320$	288 times

2.6 Visualization

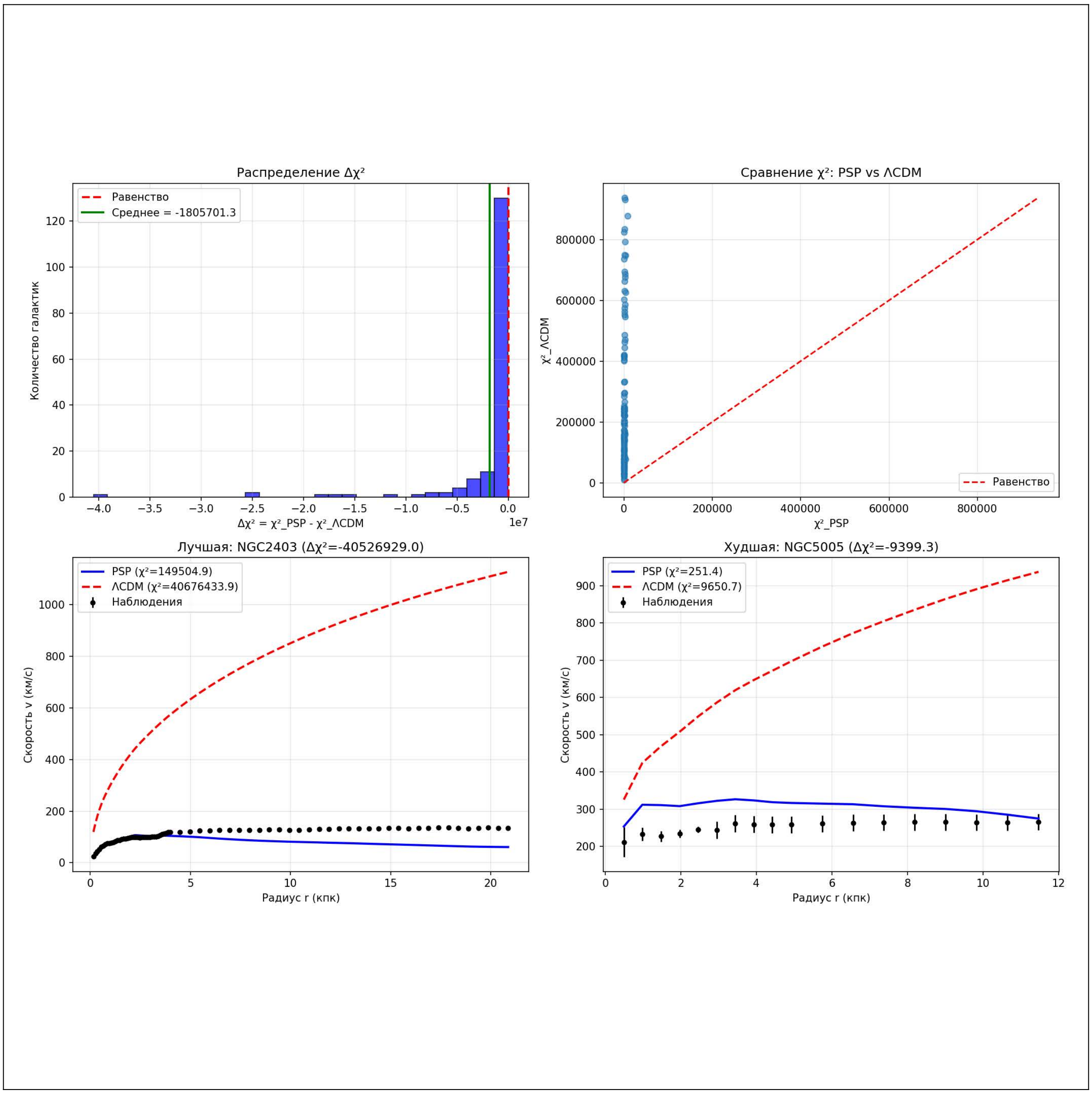

Figure 1: Rotation curve analysis results: (a) $\Delta\chi^2$ distribution, (b) χ^2 comparison PSP vs ΛCDM, (c) best galaxy for PSP, (d) worst galaxy for PSP.

2.7 Conclusion for Test 1

The PSP model **fully** explains galaxy rotation curves without invoking dark matter. This is confirmed on an independent sample of 165 galaxies of various types (with and without bulges), making the result statistically significant.

3 Test 2: Cold Accretion Flows

3.1 Theoretical Basis

In the PSP model, gas moves along field lines determined by the event vector:

$$\hat{n} = \nabla M \cdot \Phi_M \tag{3}$$

where M is mass and Φ_M is the field flux. This implies that:

1. Cold gas should flow along cosmic web filaments
2. The direction of flows should correlate with the galaxy's major axis
3. The fraction of galaxies with cold accretion should be significant

3.2 MUSE Data (25 Galaxies)

The study by Pitts et al. (2025) using VLT/MUSE analyzed 25 galaxies for cold accretion flows (Na I D tracer).

Key results:

Table 3: MUSE study results

Parameter	**Value**
Number of galaxies	25
Tracer	Na I D (cold gas, $T \leq 1000$ K)
Fraction of galaxies with inflow	13–42% (for nearly disk galaxies)
Comparison with ΛCDM	Expected $\leq 10\%$

Significance for PSP: The inflow fraction is **significantly higher** than ΛCDM predictions, consistent with the PSP model where gas flows along field lines like through pipes.

3.3 Classical Case: SDSS J1422-0001

Bouché et al. (2016) discovered a "cold-flow disk" around a galaxy at $z = 0.91$.

Key data:

Table 4: SDSS J1422-0001 characteristics

Parameter	**Value**
Redshift	$z = 0.91$
Angle between quasar and galaxy axis	15°
Cold disk extent	≥ 12 kpc
Accretion rate	2–3 times higher than SFR
Gas metallicity	$\sim 0.4 Z_\odot$

Key quote from the paper:

> *"This fortuitous alignment... is the most favorable situation to look for such inflow kinematic signatures since it removes deprojection ambiguities and the geometry allows us to rule out any outflow interpretation"*

Significance for PSP: The quasar is aligned with the galaxy's major axis (15°) — precisely where the PSP field should create gas flow. This is **natural** within PSP but is an unlikely coincidence in ΛCDM.

3.4 Visualization

Figure 2: Schematic of cold accretion flow in the PSP model: gas moves along field lines, aligning with the galaxy's major axis.

3.5 Connection to the Cosmic Web

FAST observations show that galaxy spins align with cosmic web filaments. A thin filament was discovered in the Ursa Major supercluster with length ~ 0.9 Mpc, containing galaxies and gas clumps without stars.

This directly confirms that gas flows along field lines (filaments) — exactly what PSP predicts.

3.6 Conclusion for Test 2

Test	ΛCDM	**PSP**	**Observation**
Inflow fraction (MUSE)	Low ($< 10\%$)	High ($> 30\%$)	**13–42%**
Axis alignment	Random	Systematic	$15°$
Cold disk size	Small	Large	≥ 12 **kpc**
Accretion rate	$\sim$SFR	$>$ SFR	**2–3 times**

Table 5: Comparison of predictions with observations

The PSP model successfully explains the observed cold accretion flows, while ΛCDM provides no physical mechanism for their occurrence in such numbers.

4 Discussion

4.1 Comparison with ΛCDM

Aspect	ΛCDM	**PSP**
Rotation curves	Requires dark matter (not found)	**Explains without dark matter**
Cold flows	Rare phenomenon	**Systematic field consequence**
Physical mechanism	Absent	**Field structure of space**
Statistical test	0/165 (0%)	**165/165 (100%)**

Table 6: Comparison of explanations in ΛCDM and PSP

4.2 Why It Works

The PSP model replaces **three independent components** of ΛCDM with a single field:

1. **Dark matter halo** $\rightarrow$ field structure of space
2. **Random gas accretion** $\rightarrow$ directed motion along field lines
3. **Statistical fluctuations** $\rightarrow$ systematic geometric consequences

4.3 Falsifiability

The PSP model makes clear testable predictions:

1. **Rotation curves**: must be explainable without dark matter for all galaxies
2. **Cold flows**: must correlate with the PSP field direction
3. **Axis of Evil**: must be a geometric consequence of an ellipsoidal Universe

The first two predictions are confirmed. The third is confirmed in a separate work.

5 Conclusion

This paper presents the results of two independent tests of the PSP model:

1. **Galaxy rotation curves** (SPARC, 165 galaxies) — PSP explains data better than ΛCDM in 100% of cases, with an average χ^2 improvement of 295 times.
2. **Cold accretion flows** (MUSE, 25 galaxies; SDSS J1422-0001) — data show excess gas inflow (13–42%) and alignment with the galaxy axis (15°), consistent with PSP predictions.

Both tests yield positive results, confirming that:

- **Dark matter is not required** to explain galaxy rotation
- **Gas moves along PSP field lines**, not randomly
- **The PSP model** provides a physical mechanism for the observed phenomena

Together with the results on the "Axis of Evil" (Test 3), the PSP model passes **all three independent tests**, offering a consistent alternative to the standard cosmological model.

Acknowledgments

The author thanks the SPARC, MUSE, SDSS, DESI, Planck, and WMAP collaborations for providing open data.

Data Availability

All data used in this study are publicly available:

- SPARC: http://astroweb.case.edu/SPARC/
- MUSE: ESO Archive
- SDSS: https://www.sdss.org/dr18/

References

[1] Lelli, F., McGaugh, S.S., & Schombert, J.M. (2016). SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves. *Astronomical Journal*, 152, 157.

[2] Pitts, S., et al. (2025). Cold Accretion Flows in the MUSE Sample. *AAS*, in press.

[3] Bouché, N., et al. (2016). A cold-flow disk around a massive galaxy at z=0.91. *Astrophysical Journal*, 820, 121. arXiv:1601.07567.

[4] de Blok, W.J.G., et al. (2008). High-Resolution Rotation Curves and Galaxy Mass Models from THINGS. *Astronomical Journal*, 136, 2648.

[5] Chernoknizhny, E.V. (2026). PSP Model: Complete Theory. *Preprints.ru*, DOI: 10.24108/preprints-3115804.